Algoritmos de Ordenação em tempo linear

Estruturas de Dados

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Ordenação por comparação

Qual o melhor tempo dos algoritmos de ordenação vistos até agora?

É possível ser mais rápido usando ordenação por comparação? Por quê?

É possível ordenar sem usar comparações? Em que tempo?

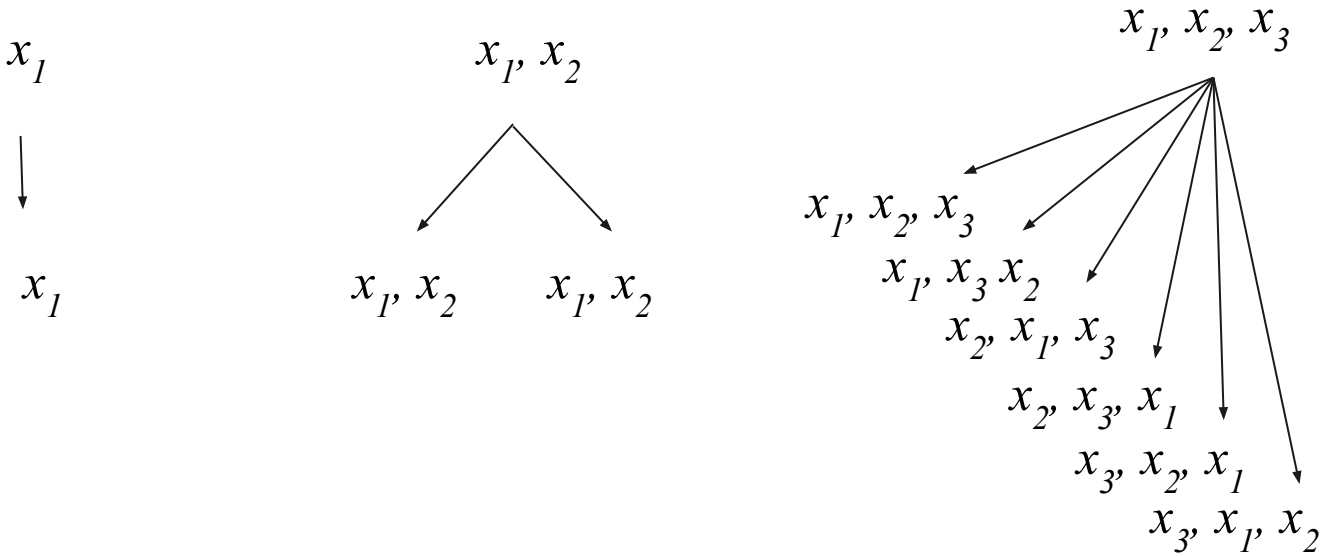
Ordenação por comparação

x_1

x_1, x_2

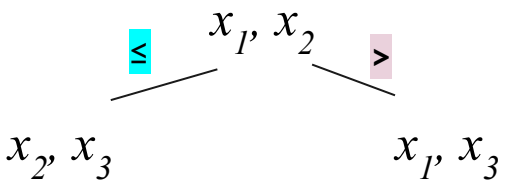
x_1, x_2, x_3

Ordenação por comparação



Ordenação por comparação (árvore de decisão)

x_1, x_2, x_3



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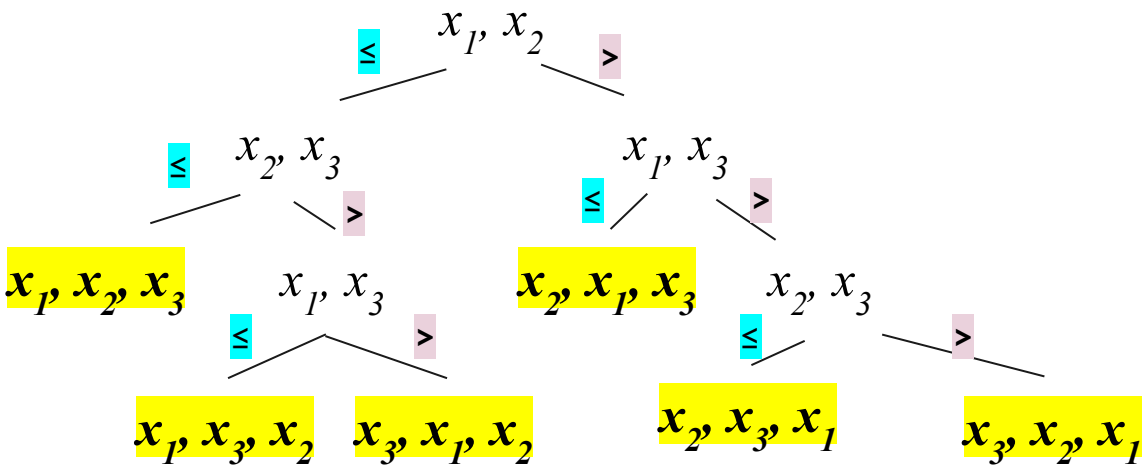

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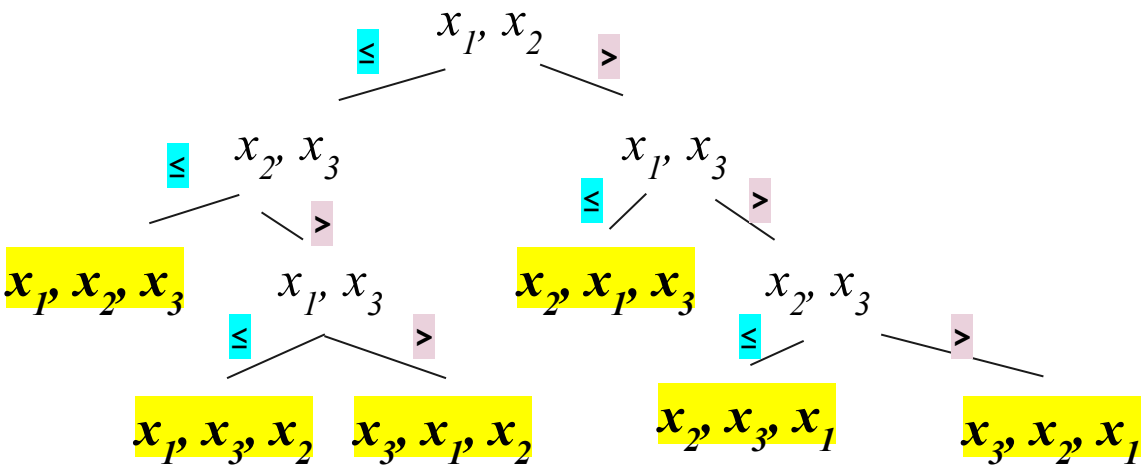

$$\#folhas \geq n!$$
$$\#folhas = 2^h$$

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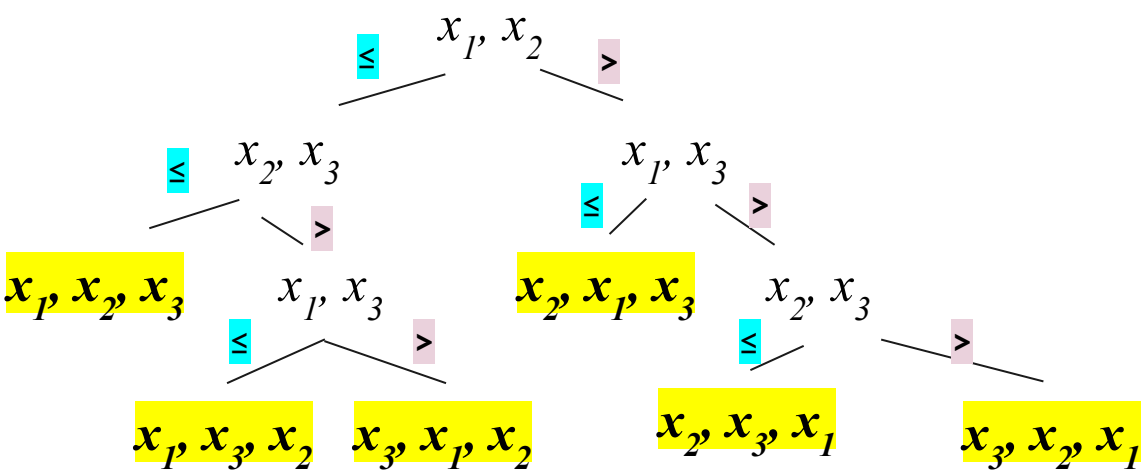

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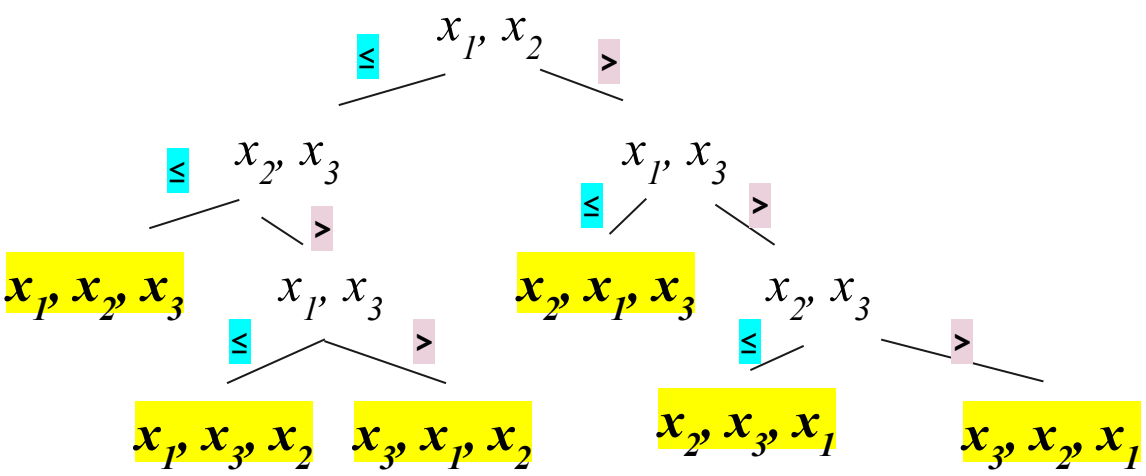

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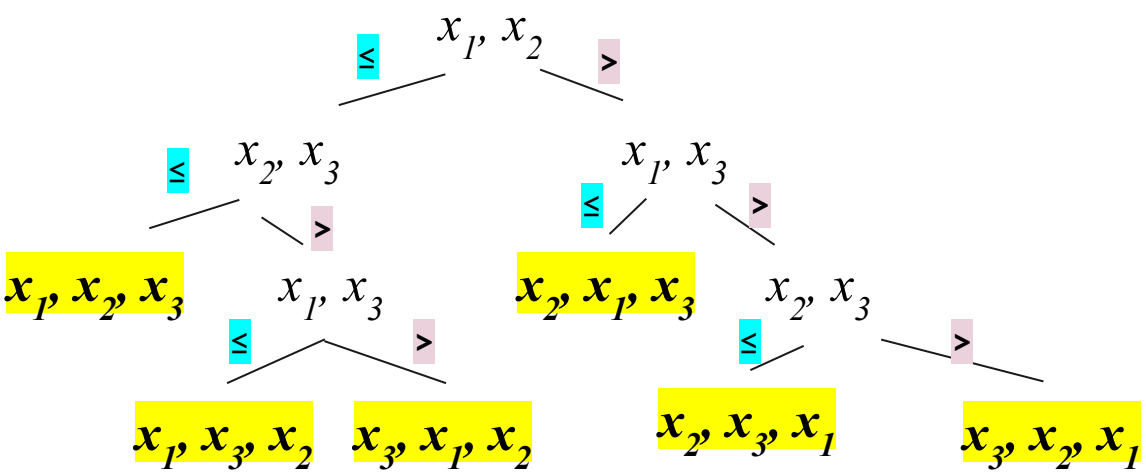

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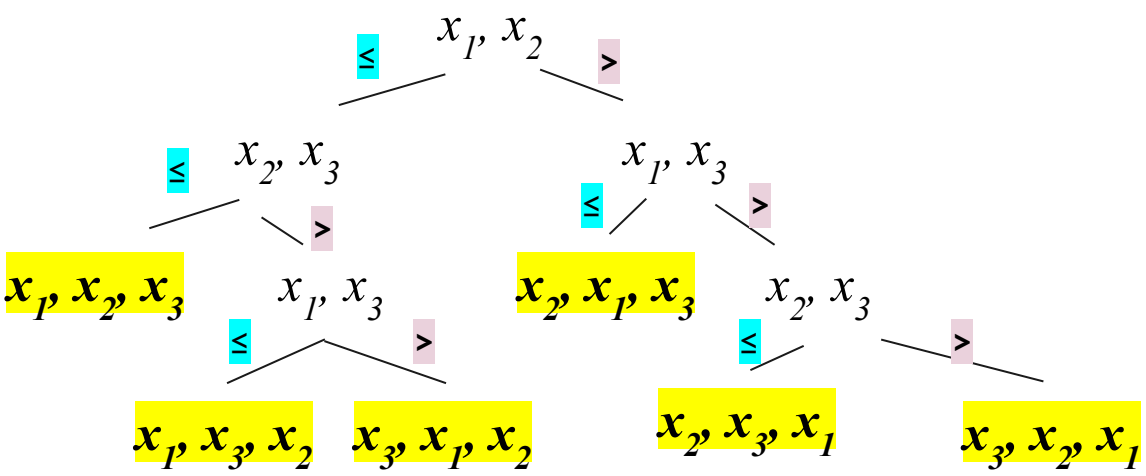

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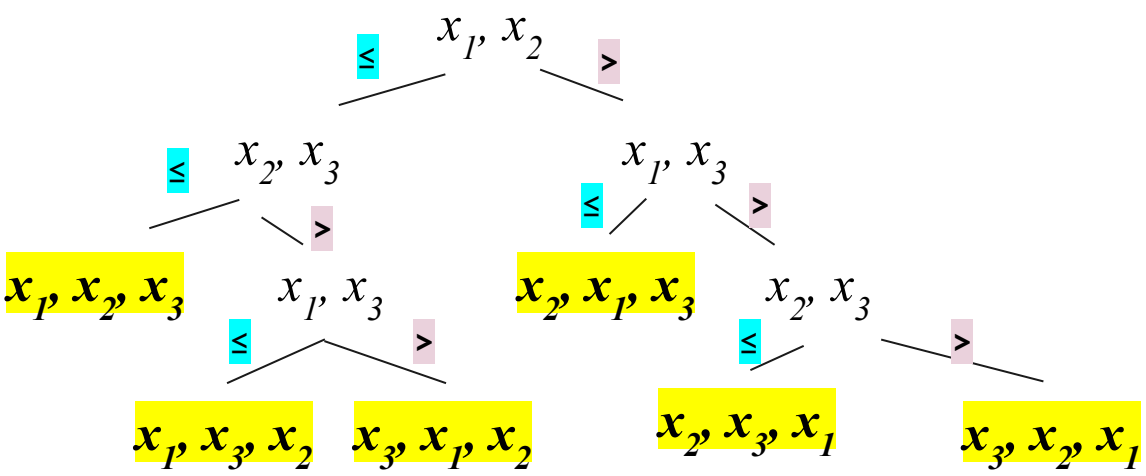

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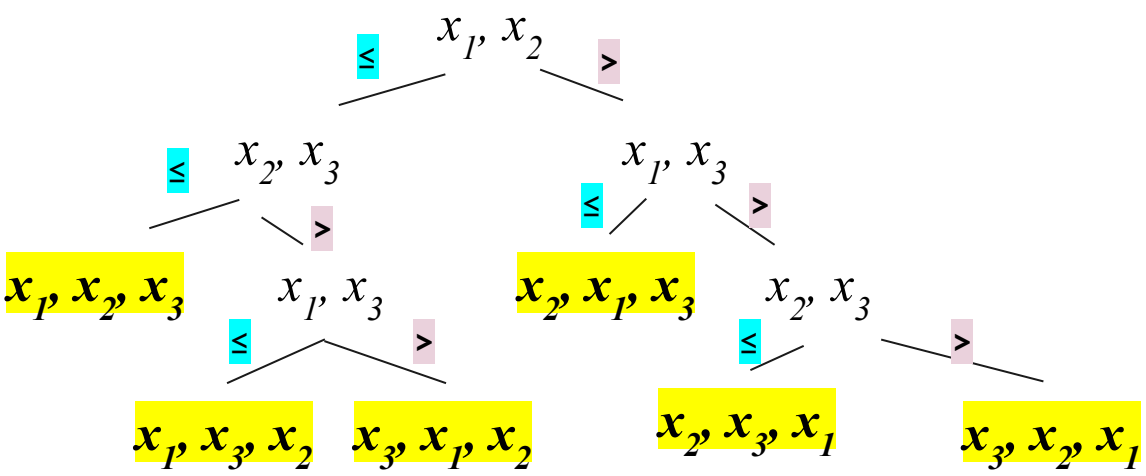

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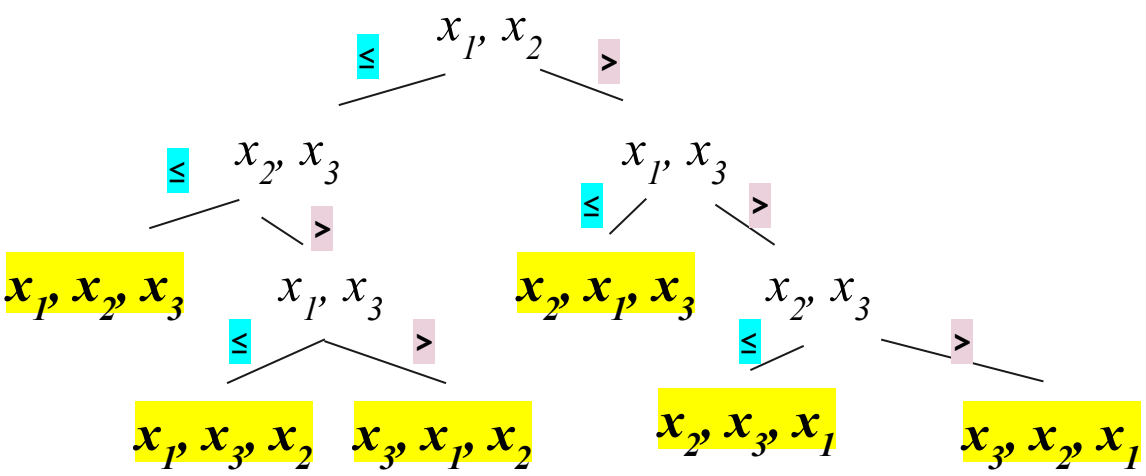

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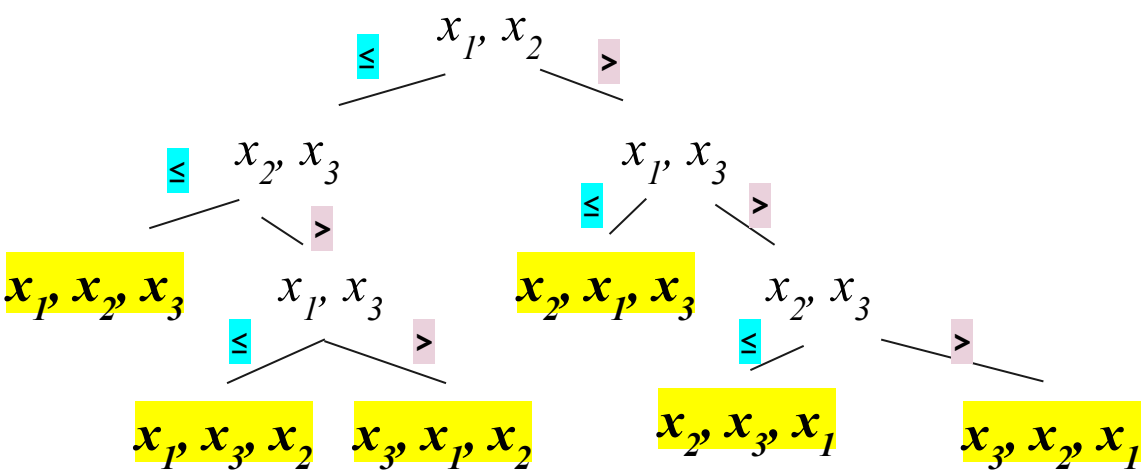

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$$h \geq n \log n$$
$$\Rightarrow h = \Omega(n \log n)$$

Ordenação sem comparações

Counting sort

Radix sort

Bucket sort

Counting sort

Counting sort

Contagem de frequências

Uso de memória extra

Saber faixa de valores a serem ordenados (1..k)

37, 825, 3, 248, ...

Counting sort

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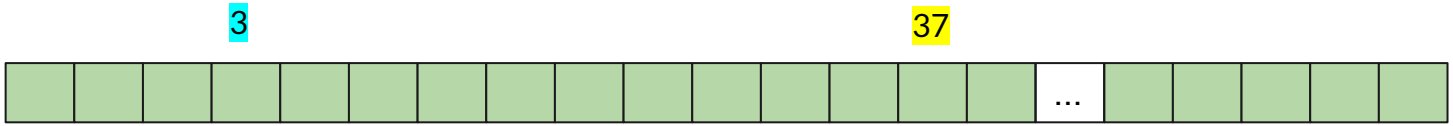

Counting sort (animação)

Counting sort (algoritmo)

[2, 3, 8, 7, 1, 2, 2, 2, 7, 3, 9]

```
COUNTING-SORT(A, B, k)
for i ← 1 to k
    do C[i] ← 0
for j ← 1 to length[A]
    do C[A[j]] ← C[A[j]] + 1
▷ C[i] now contains the number of elements equal to i.
for i ← 2 to k
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for j ← length[A] downto 1
    do B[C[A[j]]] ← A[j]
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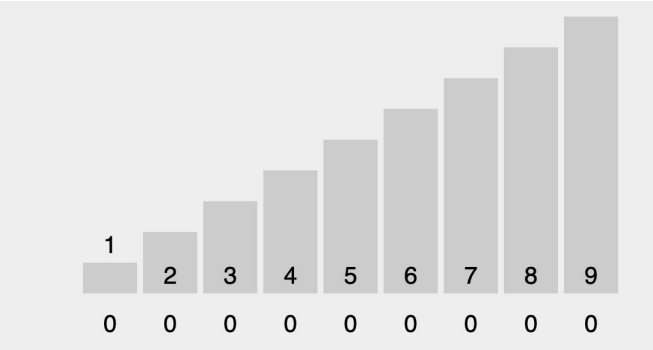


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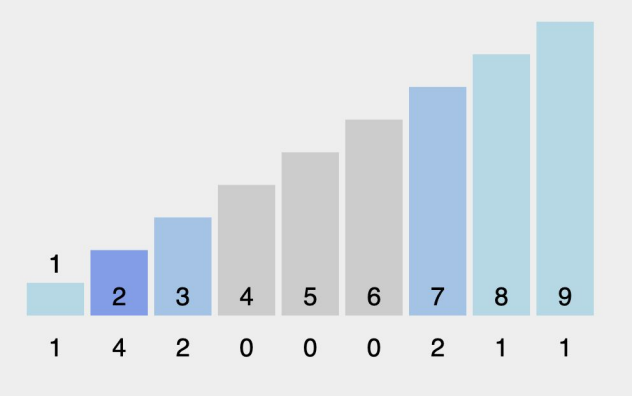


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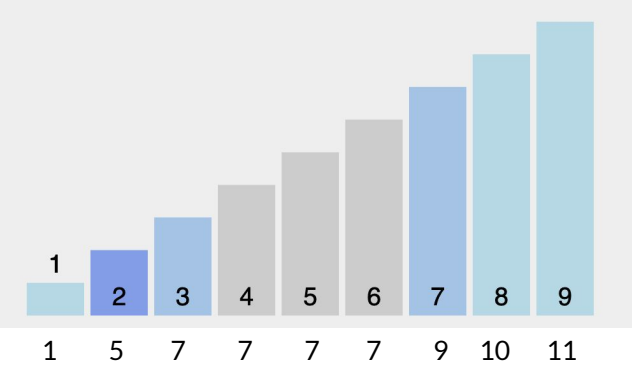


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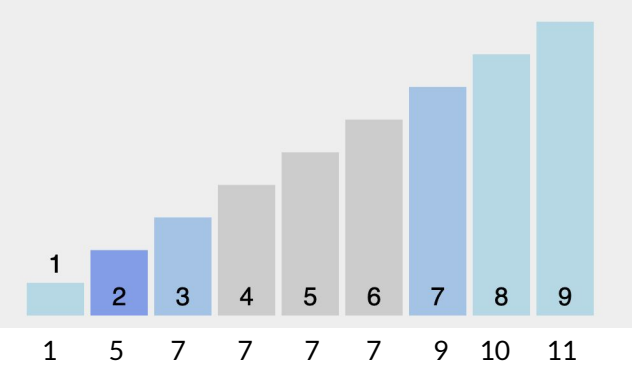

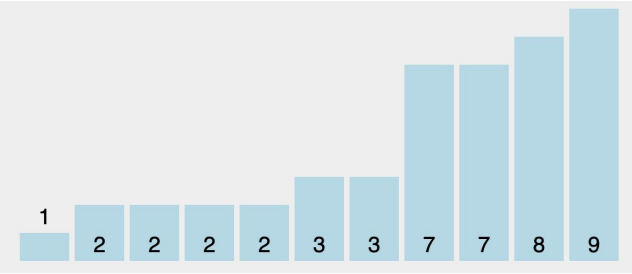

Counting sort (execução)

https://youtu.be/3bm7NgKJpj4

Counting sort (análise de tempo)

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$\boldsymbol{\theta}(k)$

Counting sort (análise de tempo)

```
COUNTING-SORT(A, B, k)
for i ← 1 to k                                          θ(k)
    do C[i] ← 0
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```

$\boldsymbol{\theta}(n + k)$

Counting sort (análise)

Estável?

In-place?

E se k >> n?

Counting sort (otimizações)

E se soubermos o mínimo e o máximo no array?

Pode-se aplicar a entradas com números negativos?

$$[x_1, x_2, x_{min}, \dots, x_k, x_{max}, \dots, x_m]$$

Counting sort (otimizações)

E se soubermos o mínimo e o máximo no array?

Pode-se aplicar a entradas com números negativos?

[10000, 10001, ... , 10010] (10 elementos)

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[10000, 10001, ... , 10010] (10 elementos)

10010 posições!!!!	10

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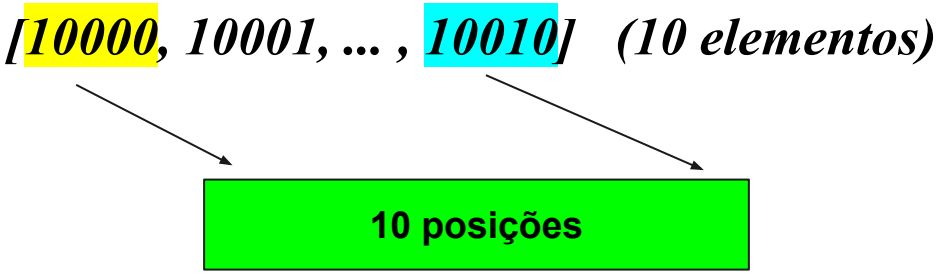

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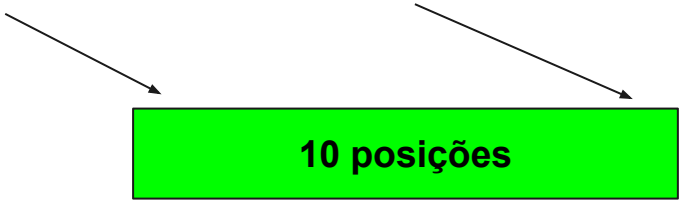


diff = 10010 - 10000 = **10**

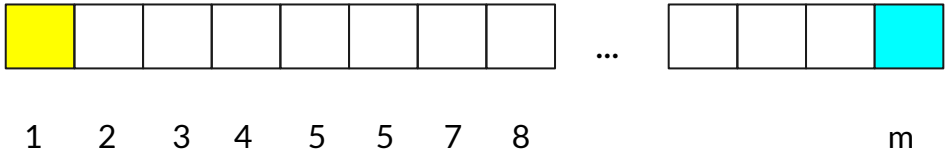
indice(x) = x - 10000

Counting sort (otimizações)

E se soubermos o mínimo e o máximo no array?

Pode-se aplicar a entradas com números negativos?

$$[x_1, x_2, x_{min}, \dots, x_k, x_{max}, \dots, x_m]$$



diff = $x_{max} - x_{min} = \mathbf{m}$

indice(x) = $x - x_{min}$

Radix sort

Radix sort

Similar a ordenação coluna a coluna do excel

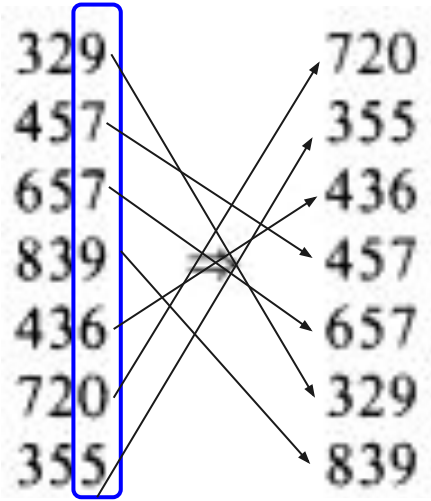
Usa ordenação dígito a dígito nos números (MSD, LSD)

Necessita de algoritmo auxiliar

Radix sort

329
457
657
839
436
720
355

Radix sort



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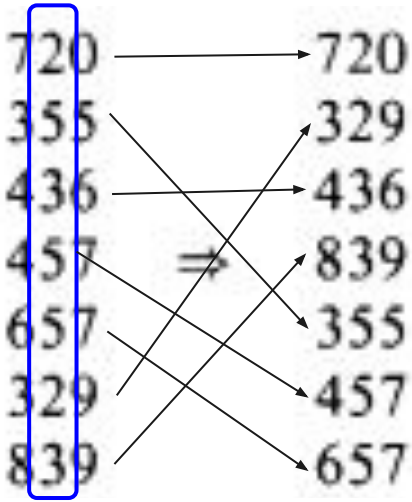

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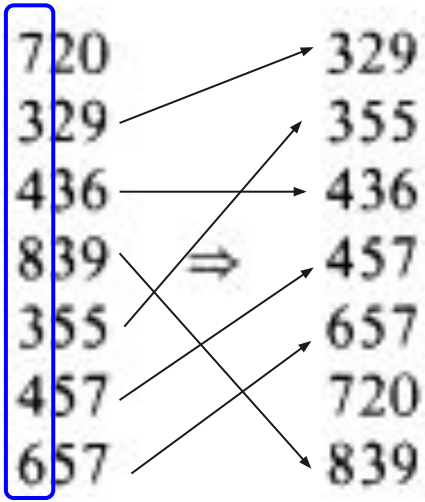


Radix sort (simulação)

Radix sort

```
radixSort( int [] A, int d) {
    for (i = 1 to d) //a partir do menos significativo
        ordene (stable) array A no dígito i
}
```

Radix sort (análise)

```
radixSort( int [] A, int d) {
   for (i = 1 to d) //a partir do menos significativo
      ordene (stable) array A no dígito i
}
```

θ (d*T(n))

Radix sort (análise)

```
radixSort( int [] A, int d) {
    for (i = 1 to d) //a partir do menos significativo
        counting sort em A no dígito i
}
```

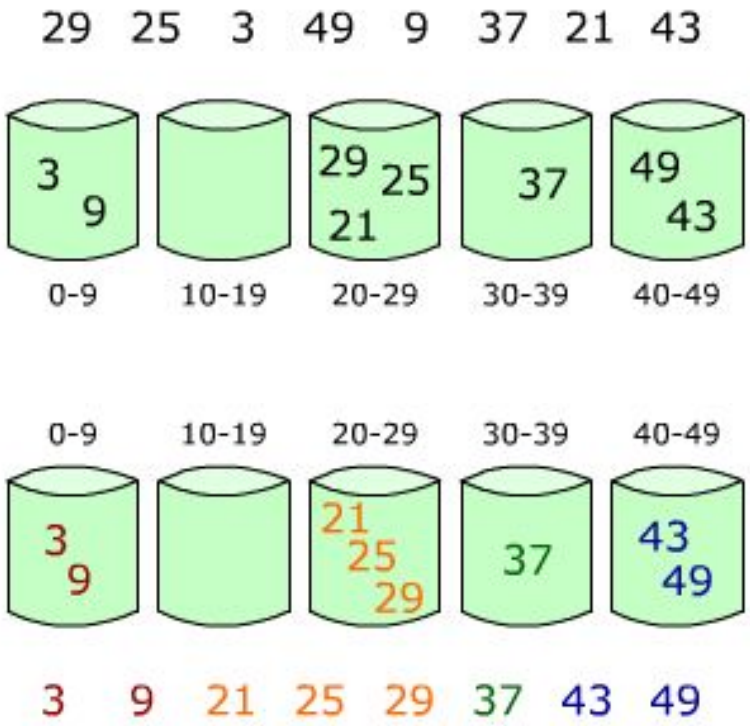
θ (d*(n + k))

⇩

θ (dn + dk)

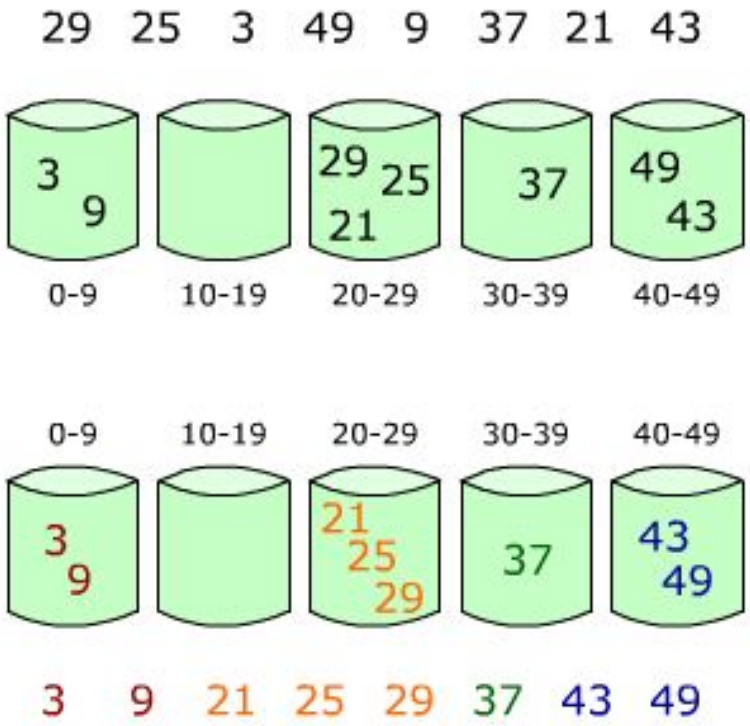
Bucket sort

Bucket sort


29 25 3 49 9 37 21 43
3 9
29 25 21
37
49 43
0-9 10-19 20-29 30-39 40-49
0-9 10-19 20-29 30-39 40-49
3 9
21 25 29
37
43 49
3 9 21 25 29 37 43 49

Bucket sort

```
function bucket-sort(array)
    buckets ← new array of 10 empty lists
    for i = 1 to array.length do
        insert array[i] into buckets[array[i]/10]
    for i = 0 to 9 do
        next-sort(buckets[i])
    return <buckets[0]>; ... ;<buckets[9]>
```



Bucket sort

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        next-sort(buckets[i])
    return <buckets[0]>; ... ;<buckets[9]>
```
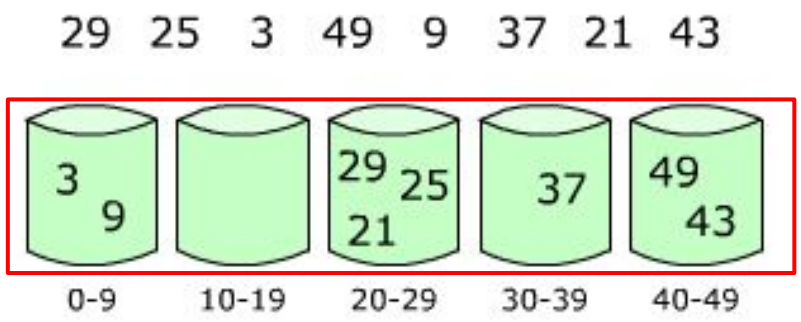
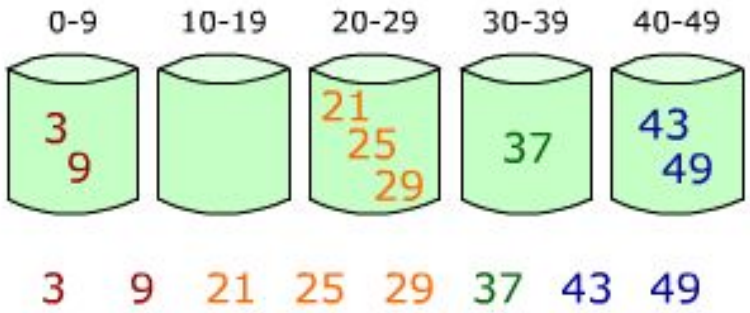



θ (n)



Bucket sort

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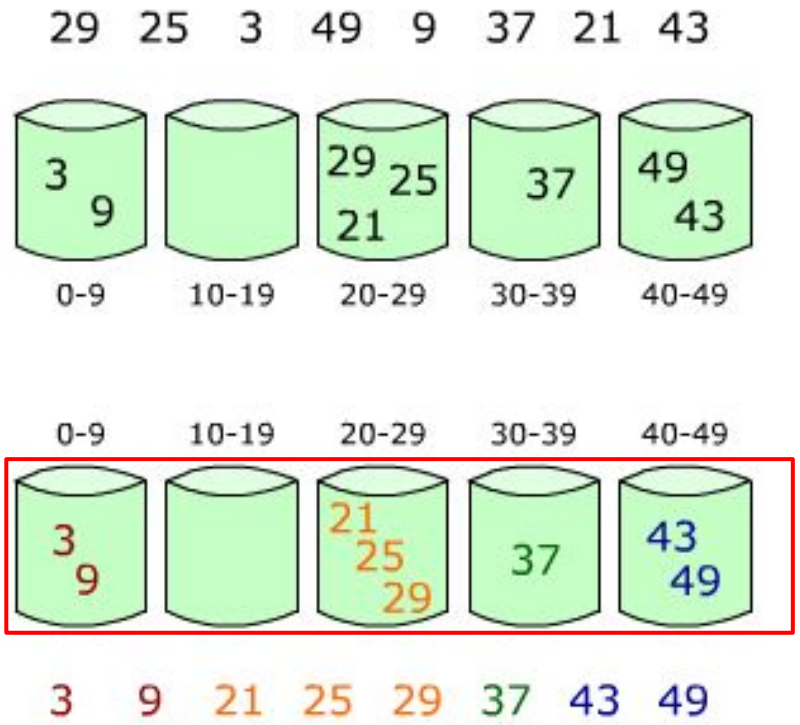
    return <buckets[0]>; ... ;<buckets[9]>
```



θ (n)

θ (b*T(n_b))

Bucket sort

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    buckets ← new array of 10 empty lists
    for i = 1 to array.length do
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```

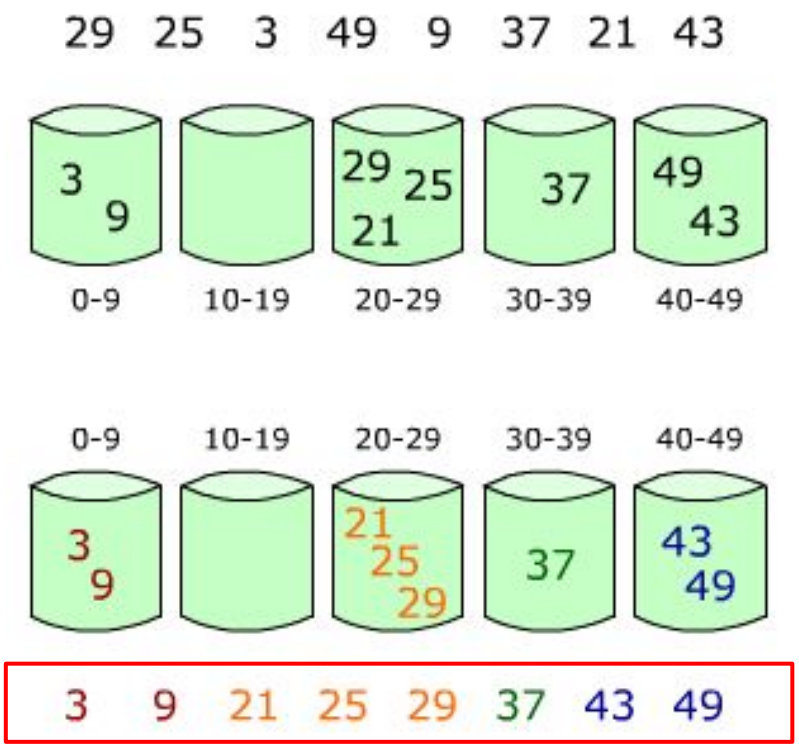

θ (n)

θ (b*T(n_b))

θ (b)

Bucket sort

```
function bucket-sort(array)
    buckets ← new array of 10 empty lists
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    return <buckets[0]>; ... ;<buckets[9]>
```

$\boldsymbol{\theta}$ (n + bf(n))

Bucket sort

```
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    for i = 1 to array.length do
        insert array[i] into buckets[array[i]/10]
    for i = 0 to 9 do
        counting sort(buckets[i])
    return <buckets[0]>; ... ;<buckets[9]>
```

$\boldsymbol{\theta}$ (n + b(n+k))

Bucket sort

```
function bucket-sort(array)
    buckets ← new array of 10 empty lists
    for i = 1 to array.length do
        insert array[i] into buckets[array[i]/10]
    for i = 0 to 9 do
        counting sort(buckets[i])
    return <buckets[0]>; ... ;<buckets[9]>
```

$$\boldsymbol{\theta}\,(n + b(n+k))$$

$$\Downarrow \quad k = \boldsymbol{\theta}\,(n),\ b << n$$

$$\boldsymbol{\theta}\,(n + n) \Rightarrow \boldsymbol{\theta}\,(n)$$